

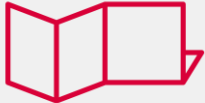


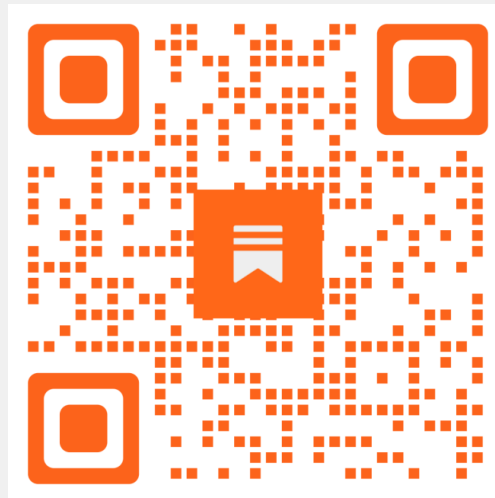
Beyond Version Control

GitHub Actions for Power BI Development

Daniel PATKOS

Speaker info

- Budapest, Hungary
- BI Architect @  VISUAL
LABS



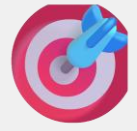


Power BI & Git - Where Are You on the Journey?

1. I develop Power BI reports — not just view them.
2. I've struggled with version control (Project_Final_v7_REAL_FINAL.pbix).
3. I already work with .PBIP projects instead of only .PBIX files.
4. I've opened or edited a TMDL file and understood (most of) what I saw.
5. I've tested my model against the Best Practice Analyzer – or I know it.
6. I use Git or another version control system for my Power BI projects.
7. I've created or reviewed a Pull Request (PR) before merging code.
8. I've seen or written a GitHub Actions or Azure DevOps YAML workflow.

Scan to refresh your Git, CI/CD, and Power BI knowledge





Main goals of this session to show how to:

- 1)  Use PBIP & Git to bring structure to Power BI development.
- 2)  Run BPA with GitHub Actions for fast, actionable feedback.
- 3)  Centralize and reuse workflows for consistent CI/CD.

Agenda

1. 🦖 Dark Age – Before version control
2. 🌱 Version control & co-op development
3. 🌿 Automated quality checks with BPA & GitHub Actions
4. 🌳 Scalable CI/CD workflows

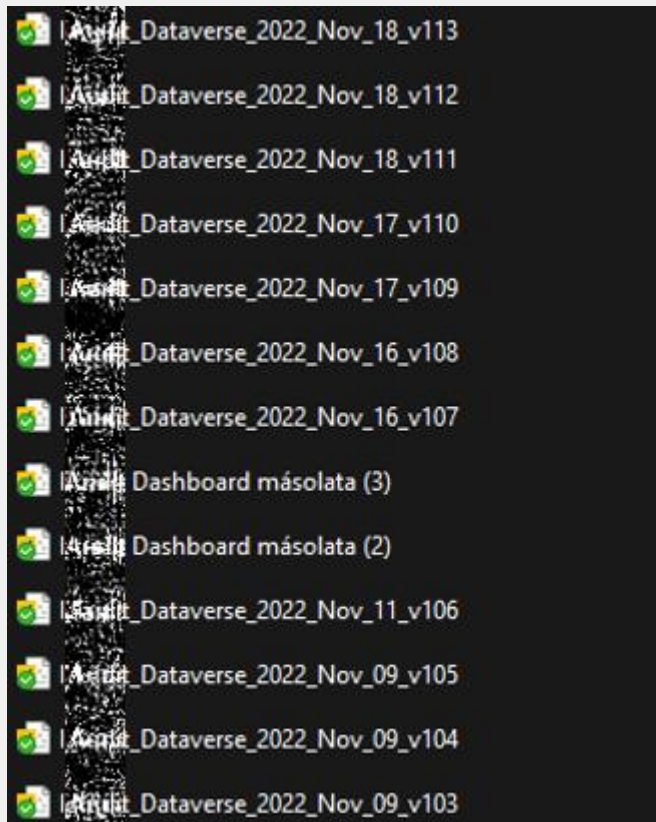


Dark Age

Before version control

The Dark Age of Power BI Version Control

Developing on your own



Developing in teams

hi, here's my version. Which one is going to be presented to the client tomorrow?

Have you uploaded your version yet? I need to add a new measure

Is your version up on sharepoint?



Version control

The start of a new era

👑 Power BI Project Files

- From .pbix to .pbip

☐ Name	Date modified
📁 AdventureWorks.Report	19/03/2024 20:43
📁 AdventureWorks.SemanticModel	19/03/2024 20:43
⚙️ .gitignore	15/03/2024 10:43
📄 AdventureWorks.pbip	18/03/2024 12:21

- Option to switch to TMDL format



- ▼ - Demo Report.Report
 - > .pbi
 - > StaticResources
 - ≡ .platform
 - ≡ definition.pbir
 - { } report.json
- ▼ - Demo Report.SemanticModel
 - > .pbi
 - ▼ definition
 - > cultures
 - ▼ tables
 - 📄 _Measures.tmdl
 - 📄 Campaign>Returns.tmdl
 - 📄 CS_NewTickets.tmdl
 - 📄 Historical_Trend.tmdl
 - 📄 Marketing_Funnel.tmdl
 - 📄 database.tmdl
 - 📄 model.tmdl
 - ≡ .platform
 - ≡ definition.pbism
 - { } diagramLayout.json
 - 🔍 .gitignore
 - 📄 Demo Report.pbip

```

1  table Historical_Trend
26  column Opened
29      lineageTag: 27c31e28-e481-4ae0-8e49-56055501a003
30      summarizeBy: sum
31      sourceColumn: Opened
32
33      changedProperty = DataType
34
35      annotation SummarizationSetBy = Automatic
36
37  column 'Converted Users'
38      dataType: int64
39      formatString: 0
40      lineageTag: d875e020-740b-4432-a433-0cc4a86eeb3b
41      summarizeBy: sum
42      sourceColumn: Converted Users
43
44      annotation SummarizationSetBy = Automatic
45
46  column 'Converted value'
47      dataType: int64
48      formatString: 0
49      lineageTag: adfbde04-734f-44e1-ba75-c1beea87094a
50      summarizeBy: sum
51      sourceColumn: Converted value
52
53      annotation SummarizationSetBy = Automatic
54
55  column 'Avg Conversion Value'
56      dataType: int64
57      formatString: 0
58      lineageTag: de3bf1d3-f77f-40fd-98a8-e9f17845a8a6
59      summarizeBy: sum
60      sourceColumn: Avg Conversion Value
61
62      annotation SummarizationSetBy = Automatic
63
64  partition Historical_Trend = m
65      mode: import
66      source =
67          let
68              Source = Table.FromRows(Json.Document(Binary.Decompress(Binary.FromText("PVPJscUwCOs1Zw5m8UItmdd/Gx8J/xzCjI0MkiDv++ggjz6jPENRwtC0x1/OT9zFc6qYp0L4R"), BinaryEncodingUtf8))
69              #"Changed Type" = Table.TransformColumnTypes(Source,{{"Converted Users", Int64.Type}, {"Converted value", Int64.Type}}),
70              #"Added Custom" = Table.AddColumn(#"Changed Type", "Avg Conversion Value", each [Converted value] / [Converted Users]),
71              #"Changed Type1" = Table.TransformColumnTypes(#"Added Custom",{{"Avg Conversion Value", Int64.Type}})
72          in
73              #"Changed Type1"
74
75      annotation PBI_ResultType = Table
76
77      annotation PBI_NavigationStepName = Navigation
78
79
    
```

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57      formatString: 0
58      lineageTag: de3bf1d3-f77f-40fd-98a8-e9f17845a8a6
59      summarizeBy: sum
60      sourceColumn: Avg Conversion Value
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62      annotation SummarizationSetBy = Automatic
63
64  partition Historical_Trend = m
65      mode: import
66      source =
67          let
68              Source = Table.FromRows(Json.Document(Binary.Decompress(Binary.FromText("PVPJscUwCOs1Zw5m8UItmdd/Gx8J/xzCjI0MkiDv++gjz6jPENRwtC0x1/OT9zFc6qYp0L4Rj"), BinaryEncodingUtf8))
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> OUTLINE

> TIMELINE

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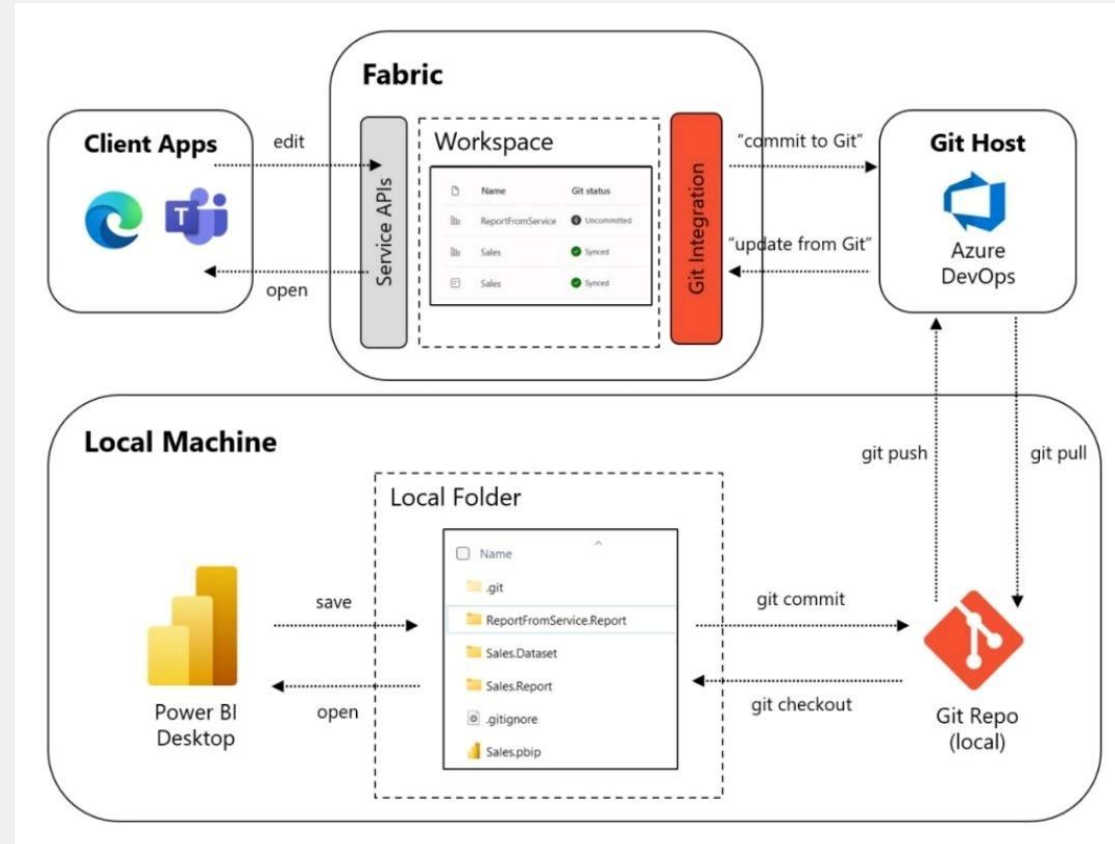
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65      mode: import
66      source =
67          let
68              Source = Table.FromRows(Json.Document(Binary.Decompress(Binary.FromText("PVPJscUwCOs1Zw5m8UIItmdd/Gx8J/xzCjI0MkiDv++gjz6jPENRwtC0x1/OT9zFc6qYp0L4R
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Git Basics



Git basics

- 1) **Repo** – stores your project related files and track all changes made to files
- 2) **Commit** – snapshot of the current status of your project
- 3) **Branch** – separate workspace wher you candevlop and try new things w/o messing up the main project
- 4) **Merge** – integrate two different versions into one
- 5) **Pull Request (PR)** – formal way to ask for approval and merge changes of 2 branches



Your next challenge: How do you...

1. Ensure consistency across multiple Power BI projects?
2. Stop suboptimal practices from leaking into production?
3. Align developers with different habits and skill levels?
4. Automate security and quality checks reliably?

Beyond version control: PBI meets CI/CD

- Apply security measure on your main branch(es)
 - Run automated tests
 - *Automate security checks and approvals*
-
- Use familiar 3rd party tools like: Tabular Editor and run:
 - Best Practice Analyzer
 - Format DAX
 - Any macros



Beyond version control: PBI meets CI/CD

BPA Severity Levels

(higher severity = greater impact)

Severity 1:  **Information:** There are Row-Level Security (RLS) roles with no members in your model.

Severity 2:  **Warning:** There are two measures with different names but defined by the same DAX expression - this should be avoided to reduce redundancy.

Severity 3:  **Error:** There are numeric columns in your model with default summarization.



Object	Type
Don't summarize numeric columns (22 objects)	
Avoid division (use DIVIDE function instead) (12 objects)	
[Internet Current Quarter Gross Profit Performance]	Measure
[Internet Current Quarter Sales Performance]	Measure
[Internet Previous Quarter Gross Profit Proportion to QTD]	Measure
[Internet Previous Quarter Sales Proportion to QTD]	Measure
[Reseller Current Quarter Sales Performance]	Measure
[Reseller Previous Quarter Gross Profit Proportion to QTD]	Measure
[Reseller Previous Quarter Sales Proportion to QTD]	Measure
[Reseller Current Quarter Gross Profit Performance]	Measure
[Total Current Quarter Gross Profit Performance]	Measure
[Total Current Quarter Sales Performance]	Measure
[Total Previous Quarter Gross Profit Proportion to QTD]	Measure
[Total Previous Quarter Sales Proportion to QTD]	Measure
Do not use floating point data types (4 objects)	
'Reseller Sales'[Discount Amount]	Column
'Reseller Sales'[Unit Price Discount Pct]	Column
'CurrencyRate'[AverageRate]	Column
'CurrencyRate'[EndOfDayRate]	Column
Avoid empty descriptions (62 objects)	
Add objects to perspectives (30 objects)	
Organize columns and hierarchies in display folders (7 objects)	
Organize measures in display folders (3 objects)	

146 objects in violation of 8 Best Practice rules.



BPA & GitHub Actions

Automated quality checks

Basic demo scenario

- 1) Someone develops a new measures and wants to push it to the *main* branch
- 2) The developer creates a Pull Request (PR)
- 3) The PR triggers a GitHub Actions workflow which will run the BPA
*Only a handful of rules are tested and only one of the rules is an error (Severity 3)

Basic demo scenario

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- 3) The PR triggers a GitHub Actions workflow which will run the BPA
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Basic demo scenario

Link to demo video



Challenge

? Run BPA checks automatically

Summary

Jobs

fetch_url

BPA

Run details

Usage

Workflow file

Annotations

1 error

BPA

failed last week in 37s

Set up job

Checkout repository

Install Tabular Editor

Run Tabular Editor BPA

1 ▶ Run ./TabularEditor.exe "PBI SUMMIT - Demo Report.SemanticModel/definition/database.tmdl" -A "./Rules_BPA.json" `

5

6 Tabular Editor 2.25.0 (build 2.25.8952.22276)

7 -----

8 Loading model...

9 Running Best Practice Analyzer...

10

11 Measure [Cost per conversion] violates rule "Avoid division (use DIVIDE function instead)"

12 =====

13 Error: Process completed with exit code 1.

Upload BPA result

Post Checkout repository

Complete job



Scaling Challenge

- ? No clear indication what must be corrected
- ? Repeated runs for multiple issues
- ✓ Run BPA checks automatically

vlpatkosdani

/ beyond-version-control-2026

<> Code

Issues

Pull requests 2

Actions

Projects

Security

Insights

Settings

Files

Demo_02_CustomMa...

+

Q

Go to file

> .github

> Contoso 10k.Report

> Contoso 10k.SemanticModel

▼ scripts

BPA_Rules.json

Custom_TA_Macro_for_BPA.csx

bpa_result_analysis.py

.gitignore

Contoso 10k.pbip

beyond-version-control-2026 / scripts

vlpatkosdani

Update Custom_TA_Macro_for_BPA.csx

This branch is 24 commits ahead of and 6 commits behind main .

Name
..
BPA_Rules.json
Custom_TA_Macro_for_BPA.csx
bpa_result_analysis.py



? No cle
? Repea

✓ Run BP

vlpatkosdani / beyond-version-control-2026

<> Code

Issues

Pull requests 2

Actions

Projects

Security

Insights

Settings

← Back to pull request #24

✓ Pr for demo 02 #15

Summary

All jobs

✓ Get_Latest_TabularEditor2_Download_Link

✓ BPA

Run details

Usage

Workflow file

BPA

succeeded now in 1m 38s

> ✓ Set up job

> ✓ Checkout repository

> ✓ Install Tabular Editor

> ✓ Run BPA Test Script

> ✓ Upload BPA CSV Result

> ✓ Set up Python

> ✓ Install Python dependencies

> ✓ Run BPA result analysis

> ✓ Upload BPA Analysis Artifacts

> ✓ Post Set up Python

> ✓ Post Checkout repository

> ✓ Complete job



Scaling Challenge

← Demo_02 - BPA and Custom Macro

✓ Pr for demo 02 #15

Summary

All jobs

- ✓ Get_Latest_TabularEditor2_Download_Link
- ✓ BPA

- Run details
- Usage
 - Workflow file

Triggered via pull request 6 minutes ago

vlpatkosdani synchronize #24 PR_for_Demo_02	Status	Total duration	Artifacts
	Success	1m 54s	2

BPA and Custom Macro.yml

on: pull_request



Artifacts

Produced during runtime

Name	Size	Digest
BPA-Analysis-CSV	4.32 KB	sha256:d875faf8209f55b125499be2e49681f1de48bd4f2324bc68039417b8065bbadc
BPA-CSV-Result	3.72 KB	sha256:2303c40172ee5ad4327bd94cb3080121f27a31e3fc035e900ef71c1df7f48260



Scaling Challenge

← Demo_02 - BPA and Custom Macro

✓ Pr for demo 02 #15

Summary

All jobs

✓ Get_Latest_TabularEditor2_Download_Link

✓ BPA

Run details

Usage

Workflow file

Triggered via pull request 6 minutes ago

👤 vipatkosdani synchronize #24 [PR_for_Demo_02](#)

Status

Success

Total duration

1m 54s

Artifacts

2

BPA and Custom Macro.yml

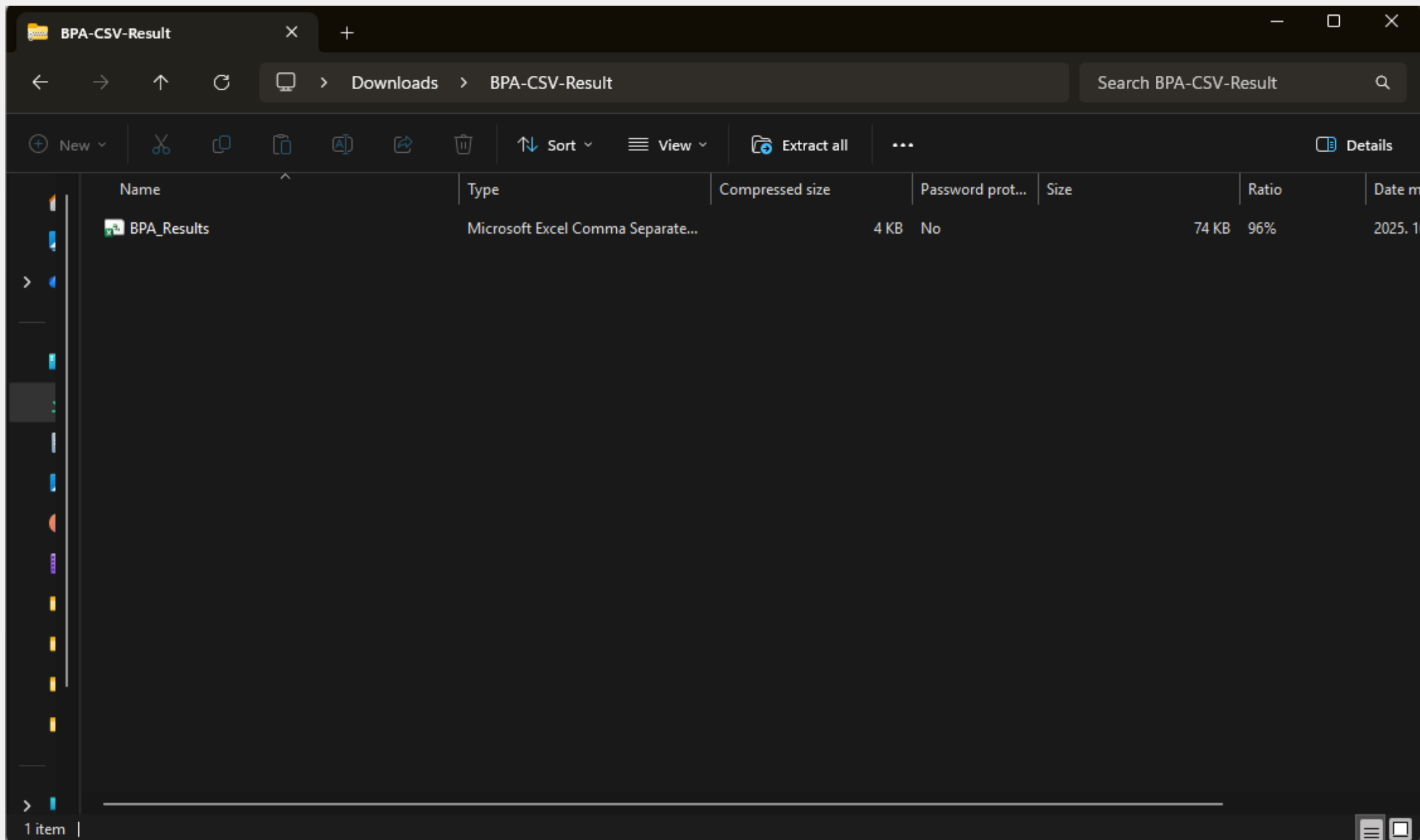
on: pull_request



Artifacts

Produced during runtime

Name	Size	Digest
📦 BPA-Analysis-CSV	4.32 KB	sha256:d875faf8209f55b125499be2e49681f1de48bd4f2324bc68039417b8065bbadc 🔗
📦 BPA-CSV-Result	3.72 KB	sha256:2303c40172ee5ad4327bd94cb3080121f27a31e3fc035e900ef71c1df7f48260 🔗



AutoSaveOff

BPA_Results - Read-Only

Search

CommentsShare

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateAdd-insHelp

GetFrom Text/CSVFrom PictureRecent Sources

From WebRecent Sources

From Table/RangeExisting Connections

Get & Transform Data

RefreshAll

Queries & ConnectionsPropertiesWorkbook Links

Queries & Connections

Stocks (En...Currencies...Geography...

SortFilterAdvanced

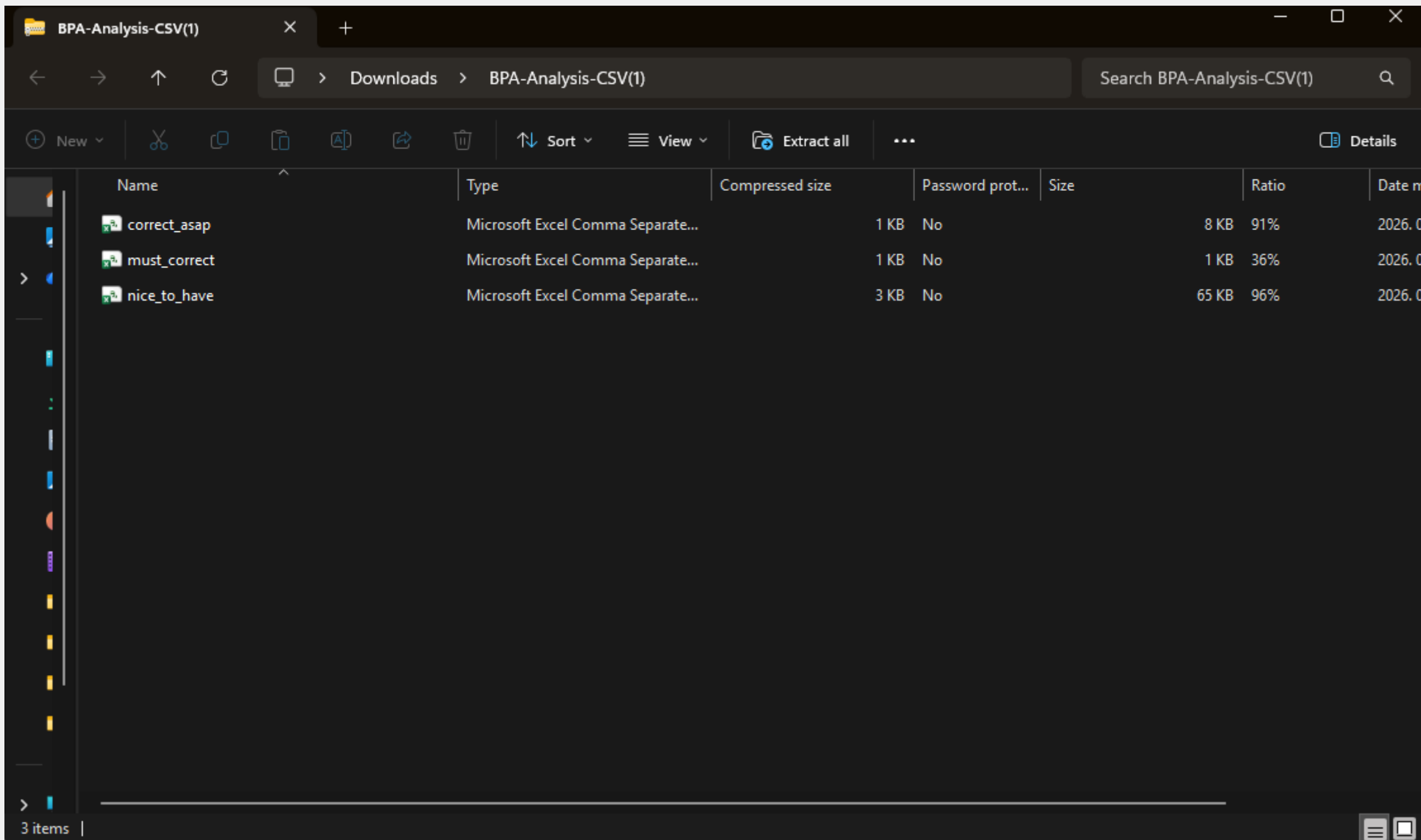
SortFilterAdvanced

Text to ColumnsClean DataFlash FillRemove DuplicatesData ValidationConsolidateData ModelAnalyze DataWhat-If AnalysisForecast SheetGroupUngroupSubtotal

Data ToolsForecastOutline

O18

	A	B	C	D	E	F	G	H	I	J	K
1	RuleCategory	RuleID	RuleName	RuleDescription	ObjectName	ObjectType	RuleSeverity	HasFixExpression			
2	Formatting	5	[Formatting] Hide fact table columns	It is a best practice to hide fact table columns that are used for aggregation in measures.		Error	2	FALSE			
3	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[CustomerKey]	Column	2	FALSE			
4	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[Address]	Column	2	FALSE			
5	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[State Code]	Column	2	FALSE			
6	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[Zip Code]	Column	2	FALSE			
7	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[Country Code]	Column	2	FALSE			
8	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Customer'[Birthday]	Column	2	FALSE			
9	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Date'[Working Day]	Column	2	FALSE			
10	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Date'[Working Day Number]	Column	2	FALSE			
11	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[ProductKey]	Column	2	FALSE			
12	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[Weight Unit Measure]	Column	2	FALSE			
13	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[Weight]	Column	2	FALSE			
14	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[Unit Cost]	Column	2	FALSE			
15	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Product'[Unit Price]	Column	2	FALSE			
16	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Order Number]	Column	2	FALSE			
17	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Line Number]	Column	2	FALSE			
18	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Order Date]	Column	2	FALSE			
19	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Delivery Date]	Column	2	FALSE			
20	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[CustomerKey]	Column	2	FALSE			
21	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[StoreKey]	Column	2	FALSE			
22	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[ProductKey]	Column	2	FALSE			
23	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Quantity]	Column	2	FALSE			
24	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Unit Price]	Column	2	FALSE			
25	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Net Price]	Column	2	FALSE			
26	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Unit Cost]	Column	2	FALSE			
27	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Currency Code]	Column	2	FALSE			
28	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Sales'[Exchange Rate]	Column	2	FALSE			
29	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Store'[StoreKey]	Column	2	FALSE			
30	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Store'[Store Code]	Column	2	FALSE			
31	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Store'[Open Date]	Column	2	FALSE			
32	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Store'[Close Date]	Column	2	FALSE			
33	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Currency Exchange'[Date]	Column	2	FALSE			
34	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Currency Exchange'[FromCurrency]	Column	2	FALSE			
35	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Currency Exchange'[ToCurrency]	Column	2	FALSE			
36	Metadata	7	Disable attribute hierarchies to decrease processing	Disable Attribute hierarchies for hidden collumns. This will ensure faster processing.	'Currency Exchange'[Exchange]	Column	2	FALSE			
37	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Customer'[CustomerKey]	Column	2	FALSE			
38	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Customer'[Address]	Column	2	FALSE			
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40	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Customer'[Zip Code]	Column	2	FALSE			
41	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Customer'[Country Code]	Column	2	FALSE			
42	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Customer'[Birthday]	Column	2	FALSE			
43	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Date'[Working Day]	Column	2	FALSE			
44	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Date'[Working Day Number]	Column	2	FALSE			
45	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Product'[ProductKey]	Column	2	FALSE			
46	Performance	8	[Performance] Set IsAvailableInMdx to false on non-attribute columns	To speed up processing time and conserve memory after processing, attribute hierarchies	'Product'[Weight Unit Measure]	Column	2	FALSE			



Demo

- ✓ No clear indication what must be corrected
- ✓ Repeated runs for multiple issues
- ✓ Run BPA checks automatically



Scaling challenges

- ? Focus on important issues first
 - ? Get additional context
 - ? Lack of detailed feedback on Pull Request
-
- ✓ No clear indication what must be corrected
 - ✓ Repeated runs for multiple issues
 - ✓ Run BPA checks automatically



vlpatkosdani / beyond-version-control-2026

<> Code

Issues

Pull requests 2

Actions

Projects

Security

Insights

Settings



Files



Demo_03_Analysis_a...



Go to file



> .github

> Contoso 10k.Report

> Contoso 10k.SemanticModel

▼ scripts

BPA_Rules.json

Custom_TA_Macro_for_BPA.csx

bpa_result_analysis.py

pr_comment.py

.gitignore

Contoso 10k.pbip

beyond-version-control-2026 / scripts /



vlpatkosdani update files

This branch is 6 commits ahead of and 4 commits behind main .

Name



..



BPA_Rules.json



Custom_TA_Macro_for_BPA.csx



bpa_result_analysis.py



pr_comment.py



❌ Model cannot be deployed until Severity 3 issues are fixed.

📁 BPA Result & Result analysis (CSV downloads): [View .zip files Here](#)

📊 Summary

Severity Level	Number of Issues
🔴 Must Correct	1
⚡ Correct ASAP	41
💡 Nice to Have	201

🔴 Must Correct (Severity 3)

RuleName	Violations
Avoid division (use DIVIDE function instead)	1

▼ 🔵 Avoid division (use DIVIDE function instead) – [Rule Description](#) 🔗

- [Rule Violation - Division] (Measure)

⚡ Correct ASAP (Severity 2)

RuleName	Violations
Disable attribute hierachies to decrease processing	34

Demo

- ✓ Lack of detailed feedback on Pull Request
 - ✓ Focus on important issues first
 - ✓ Give additional context
-
- ✓ No clear indication what must be corrected
 - ✓ Repeated runs for multiple issues
 - ✓ Run BPA checks automatically



Scalable CI/CD workflows

Answering multi-project environment's needs



Scaling challenges

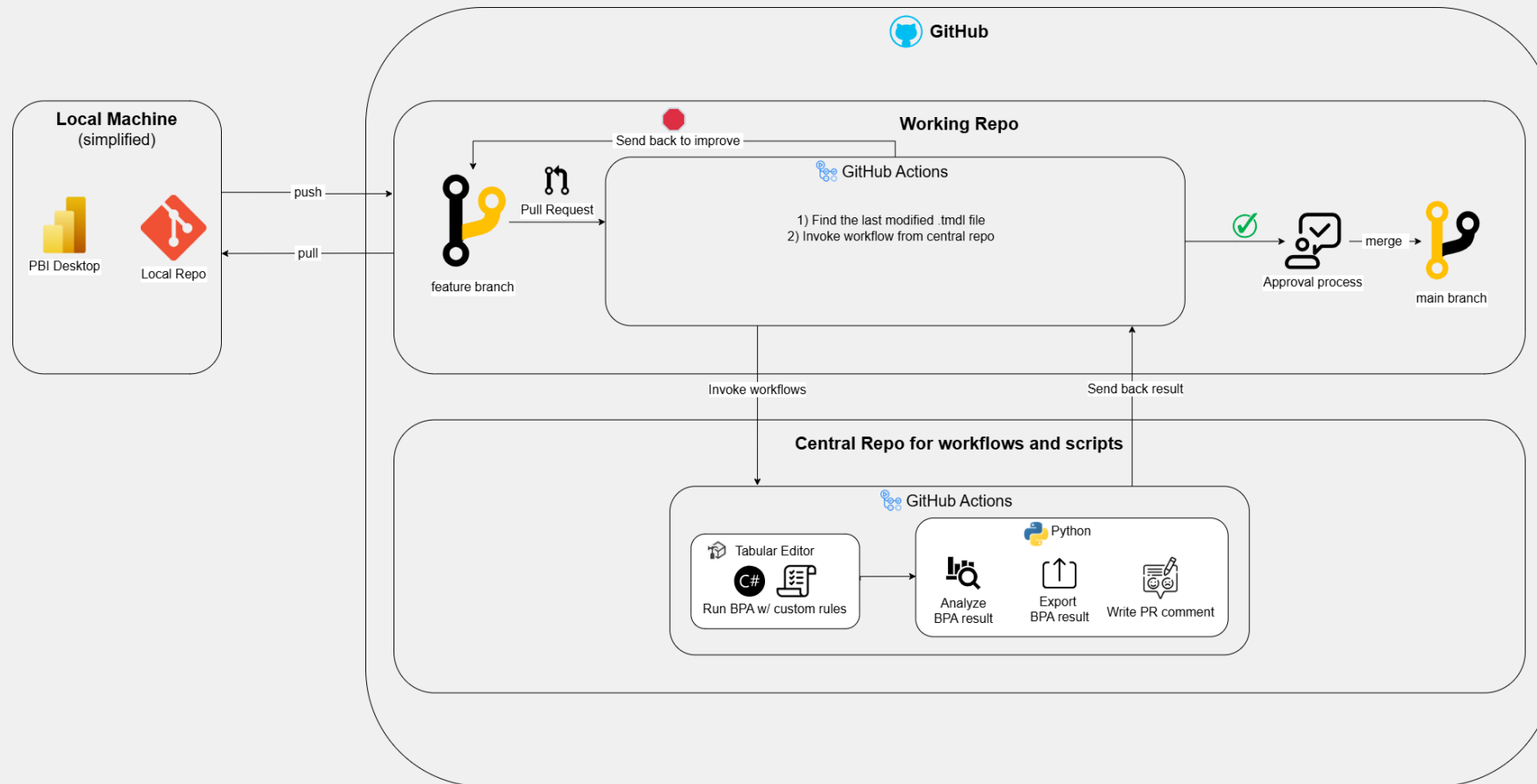
- ? Version control of the solution package
- ? Automatically create GA workflow to each newly created repository
- ? Set default approver(s)
- ? Control which part(s) of the workflow you need to run based on PR

Reusable workflow across the organization

- ✓ Version control of the solution package

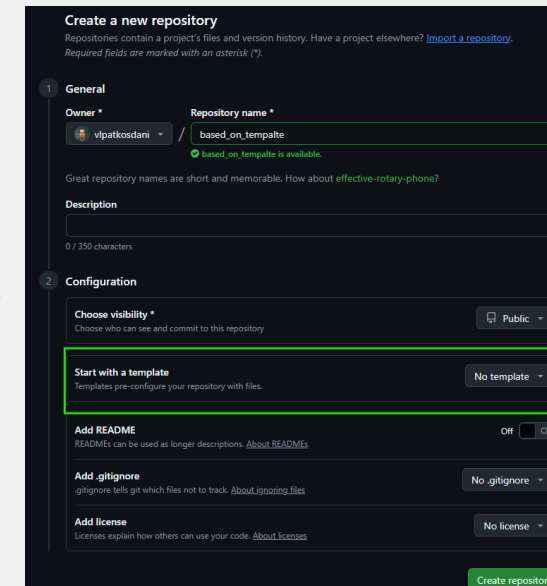
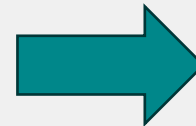
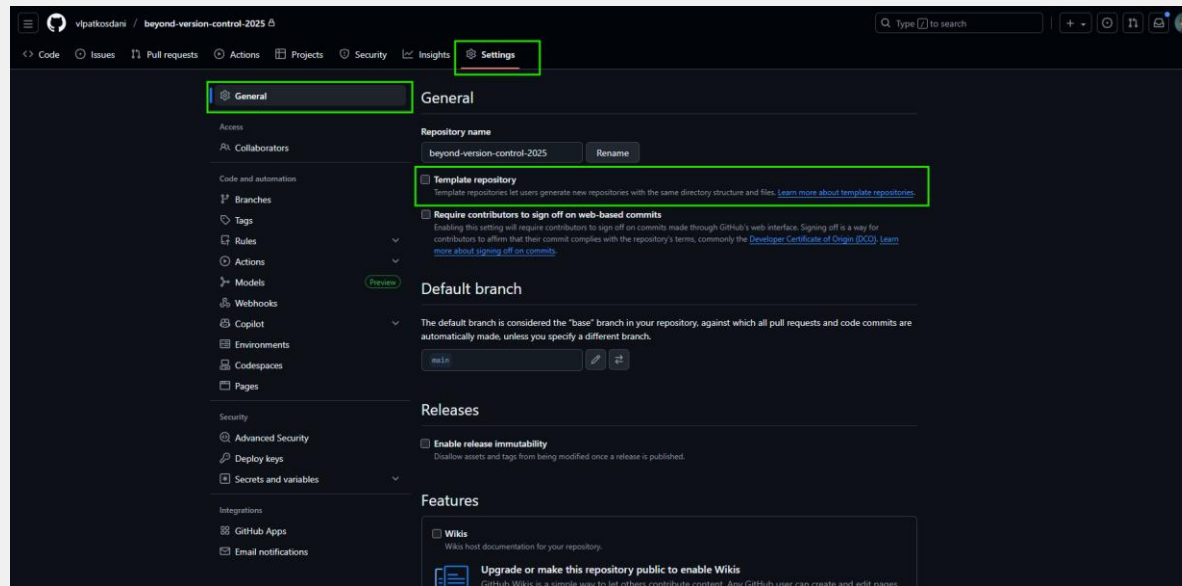
```
run-bpa:
  needs: find-latest-tmdl
  uses: Your-Organization/central-repo-for-automated-pr-checks/.github/workflows/Reusable_BPA_Checker.yml@main
  with:
    tmdl_path: ${ needs.find-latest-tmdl.outputs.tmdl_path }
  secrets:
    Central_Repo_PAT: ${ secrets.Central_Repo_PAT }
```

PR triggered BPA – stored in central repo



Template repo

✓ Add workflow for each newly created repository



Assign default approvers 1

✓ Set default approvers

```
assign_reviewer:
  needs: run-bpa
  if: github.event_name == 'pull_request'
  runs-on: ubuntu-latest
  steps:
    - name: Assign Reviewer
      run: |
        curl -X POST \
          -H "Authorization: token ${ secrets.GITHUB_TOKEN }" \
          -H "Accept: application/vnd.github.v3+json" \
          https://api.github.com/repos/${ github.repository }}/pulls/${ github.event.pull_request.number }/requested_reviewers \
          -d '{"reviewers": ["vlpatkosdani"]}'
```

Assign default approvers 2

 Set default approvers

Add CODEOWNERS file in the repo + set up branch protection rules

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).



base: Demo_03_Analysis_and_PR_Com...



compare: PR_Template

✓ **Able to merge.** These branches can be automatically merged.



Add a title

Pr template

Add a description

Write

Preview

##Context

Gives the reviewer some context about the work and why this change is being made, the WHY you are doing this. This field goes more into the product perspective.

Releted ADO Item(s)

List all related ADO items - use URLs!

Summary of changes

High level descreiption of the changes

Details

Provide a detailed description of how exactly this task will be accomplished. This can be something technical. What specific steps will be taken to achieve the goal? This should include details on service integration, job logic, implementation, etc.

Further info

- ☒ Semantic Model changes
- ☐ Visualization changes

Reviewers



Suggestions

Copilot

Request

Assignees



No one—[assign yourself](#)

Labels



None yet

Projects



None yet

Milestone



No milestone

Development

Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

[GitHub Community Guidelines](#)

Create pull request



Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

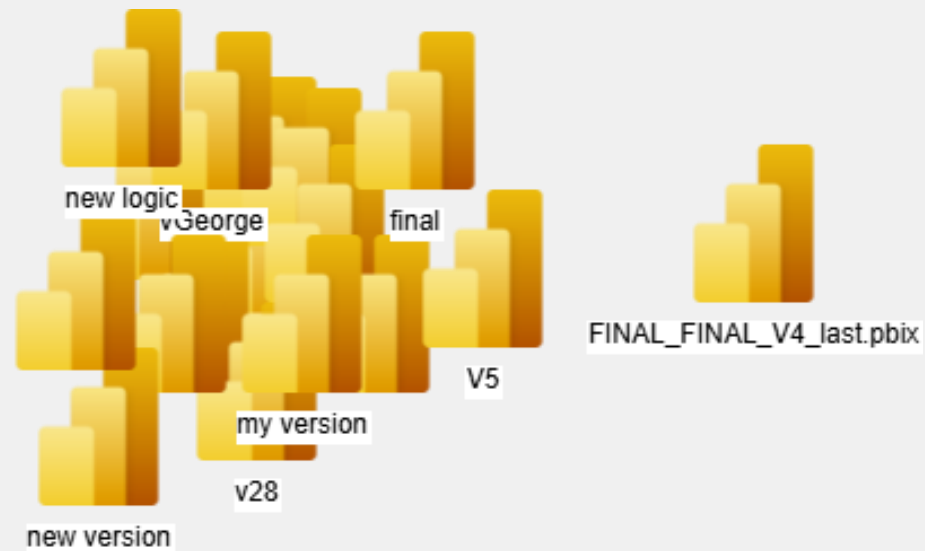
Monitor all repos

- Send logs of all runs into a central repo
- Understand the state, strength and areas of improvements of your models across the entire organization and your team

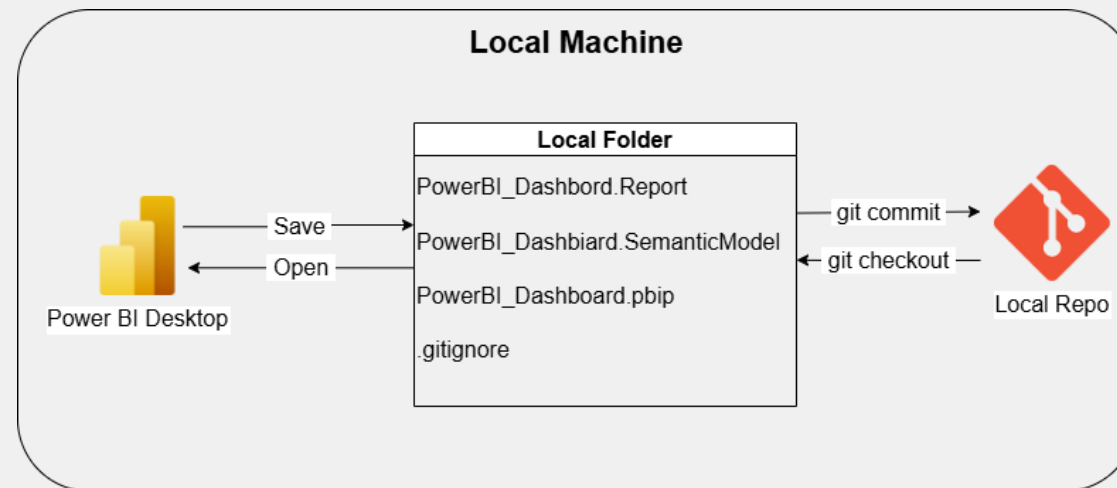


Wrap up

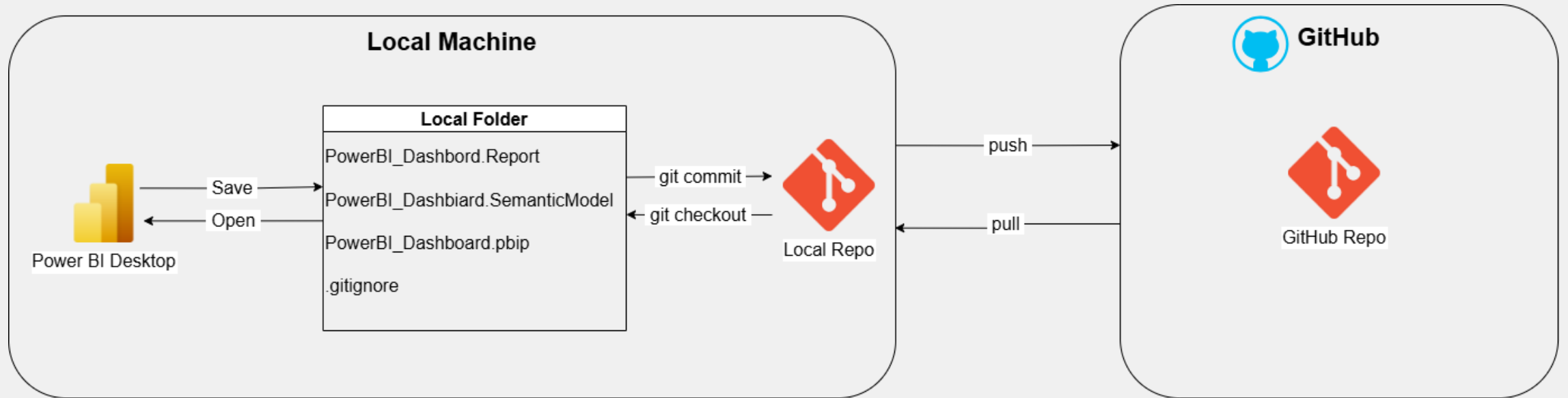
Version Chaos



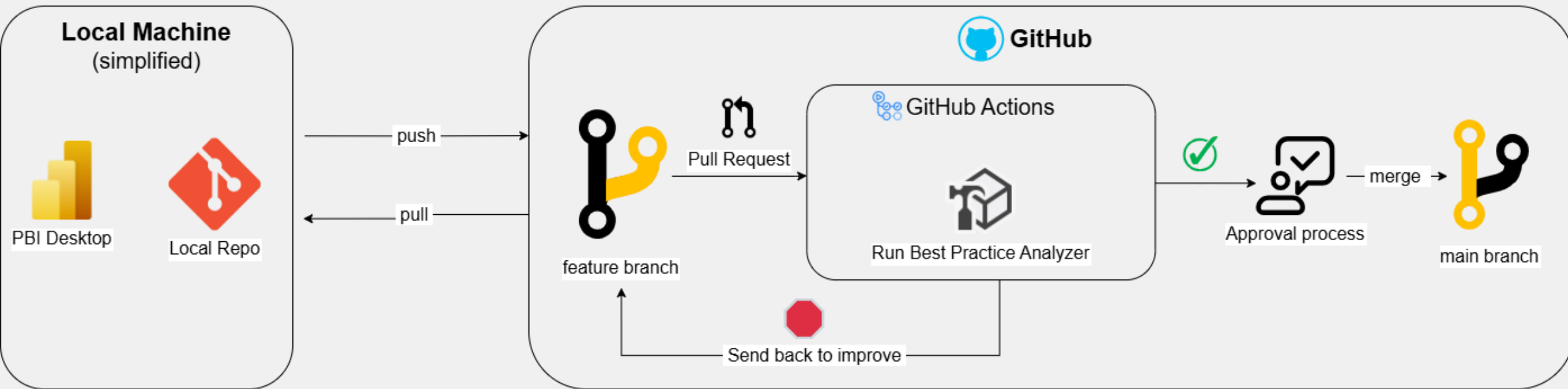
Local version control



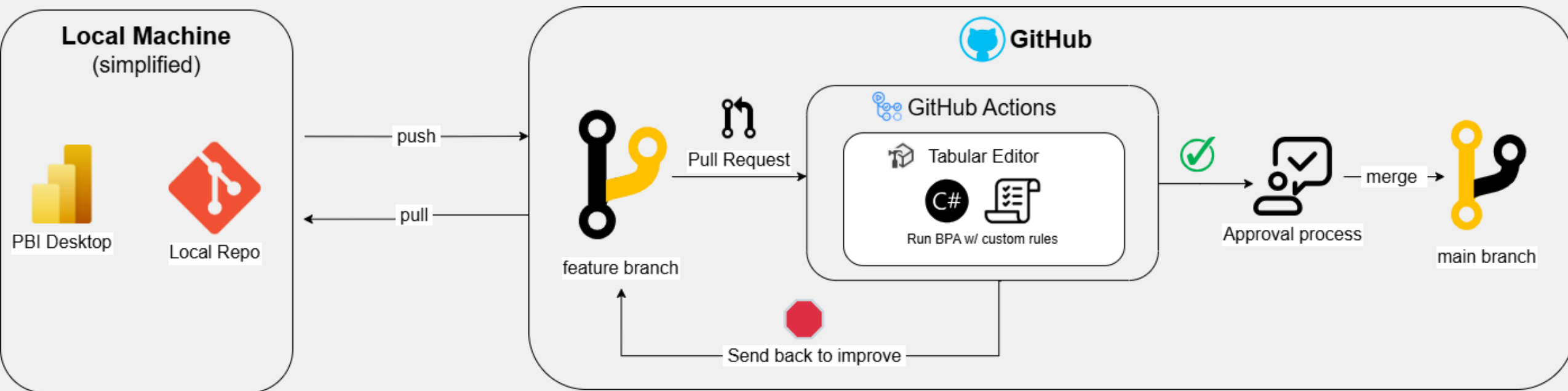
Repo stored on GitHub



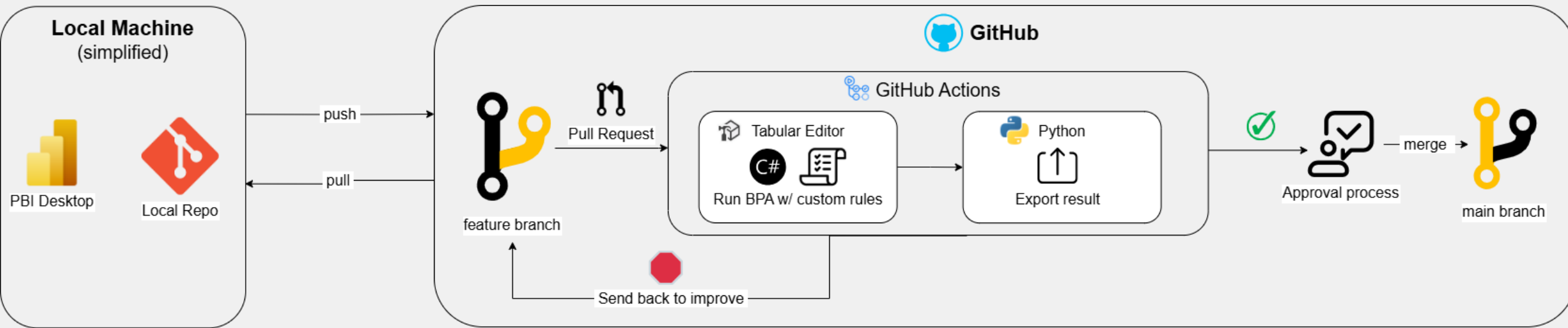
PR triggered BPA analysis



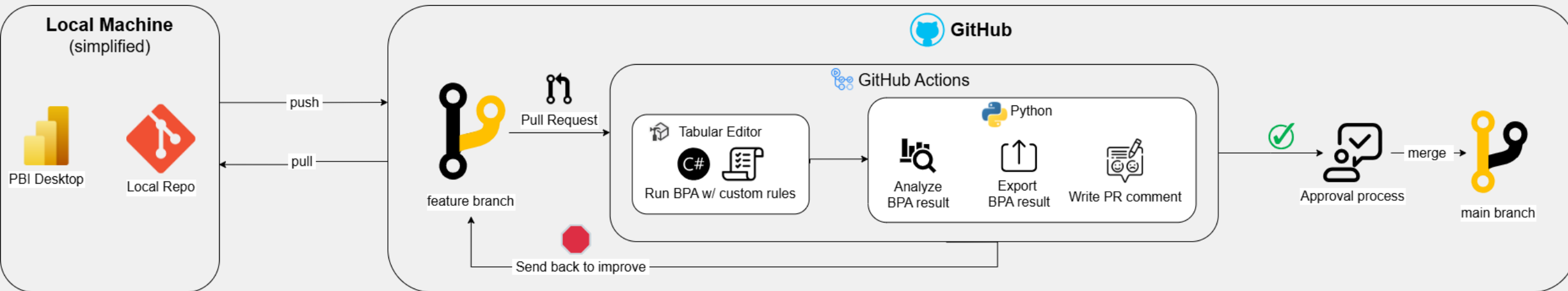
PR triggered BPA – custom rules



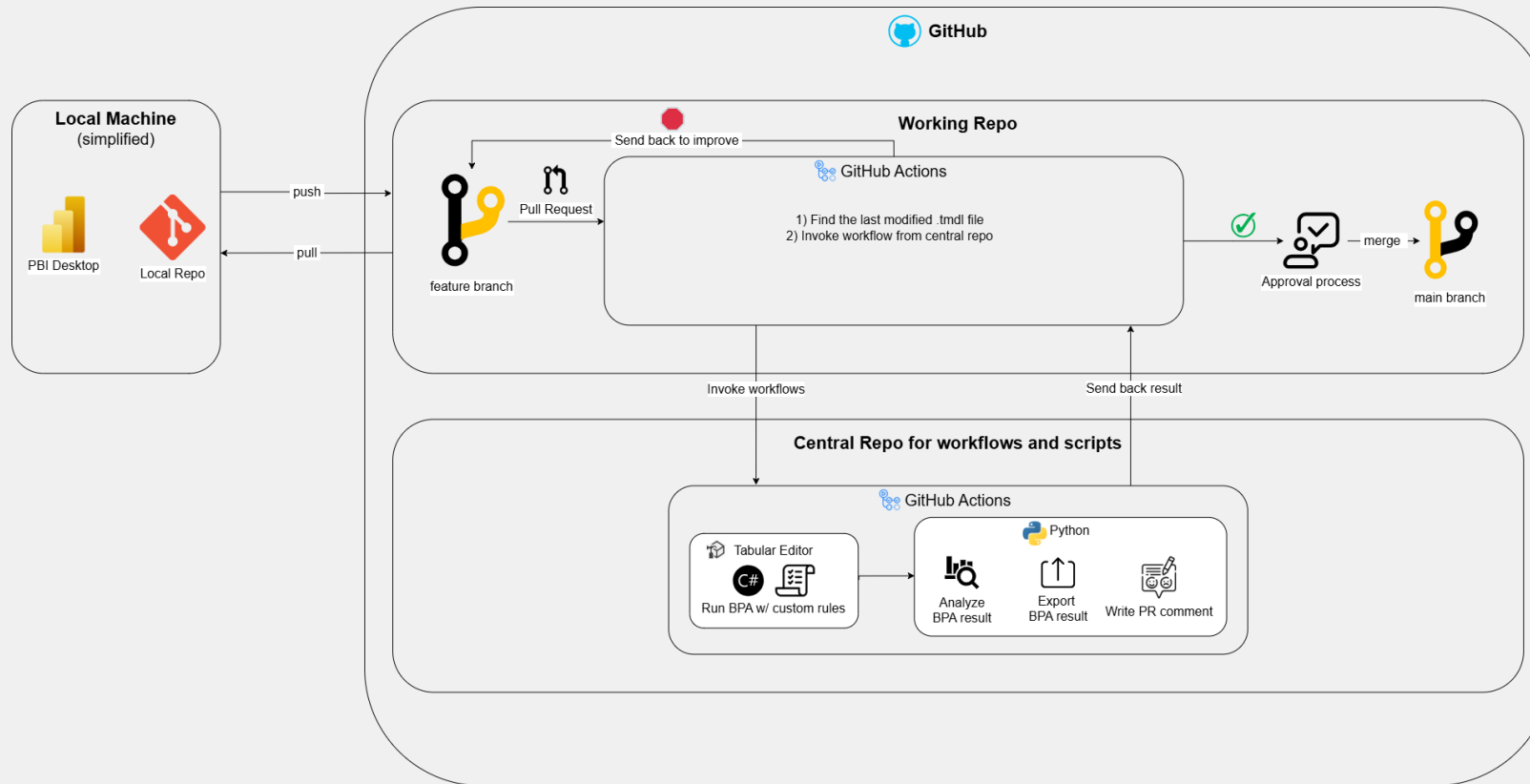
PR triggered BPA – export results



PR triggered BPA – detailed PR message



PR triggered BPA – stored in central repo



Reminder

Make sure everyone is using the way you intended and everyone is are comfortable using it

 **Start using it now!** 



Resources to study more

[Kevin Chant](#)

[Peer Grønnerup](#)

[Rui Romano](#)

[Mathias Thierbach](#)

- **Time for: thoughts, questions, or reactions**

